LMDIS

Lossless Multimodal Document Intelligence System

Graph Guided Evidence Aware Reasoning
for Enterprise Document Intelligence

Project Overview and Technical Documentation

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Repository: https://github.com/nihcastics/LMDIS-with-Graph-Guided-Evidence-Aware-Reasoning

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1. Executive Summary

LMDIS is an enterprise-grade document intelligence platform that addresses the critical challenge of extracting, structuring, and retrieving information from complex multimodal documents without data loss. The system combines lossless parsing techniques, graph-based knowledge representation, semantic retrieval, and evidence-grounded response generation to deliver traceable and accurate document analytics.

The platform ingests PDF documents in any format (digital, scanned, or hybrid), performs comprehensive content extraction preserving layout hierarchy and visual elements, constructs semantically enriched knowledge graphs, and serves natural language queries through a dual-path reasoning architecture that ensures factual accuracy through citation-backed responses.

Key Achievement: A production-ready 21-module deterministic pipeline that transforms complex multimodal documents into queryable knowledge graphs with evidence-grounded, citation-accurate answers.

2. System Architecture

2.1 Document Processing Pipeline

The document processing pipeline executes through four distinct stages:

Stage 1: Document Ingestion and Parsing (M1-M6)

Documents are registered with cryptographic fingerprinting, classified by type (digital, scanned, or mixed), and processed through lossless content extraction that preserves text, tables, images, and layout structure.

Stage 2: Semantic Processing and Graph Construction (M7-M9b)

A knowledge graph is constructed using NetworkX with 15+ edge types, enriched with semantic metadata, and indexed with 1024-dimensional embeddings for efficient retrieval.

Stage 3: Knowledge Extraction and Evidence Graph (M10-M11, LLM-K)

Claims are extracted using rule-based methods, integrated into an evidence graph with inter-claim relationships, and supplemented with LLM-derived knowledge for comprehensive query resolution.

Stage 4: Indexing and Storage

All processed data is serialized and stored in per-document isolated storage, enabling individual document management without cross-contamination.

2.2 Query and Retrieval System

The query pipeline implements a sophisticated multi-stage retrieval and dual-path answer generation:

1. **Query Interpretation (M12):** Intent detection and sub-claim decomposition
2. **Multi-Strategy Retrieval (M13-15):** Bi-encoder search, graph traversal, cross-encoder re-ranking
3. **Dual-Path Generation:** Evidence-bound answers (M16-19) and LLM Knowledge Store (Path B)
4. **Answer Selection Layer (ASL):** Multi-signal scoring and arbitration
5. **Formatting (M20):** Citation linking and confidence scoring
6. **Suggestions (M21):** Context-aware follow-up questions

3. Module Inventory

The system comprises 21 specialized modules organized into logical processing stages:

Module	Name	Function
M1	Ingestion	Immutable source registration with SHA-256 fingerprinting
M2	Page Type Classifier	Classifies pages as digital, scanned, or mixed
M3	Content Extraction	Lossless text, table, and image extraction
M4	Line Memory	Line-level normalization and paragraph detection
M5	Structure Detection	Hierarchical section and heading recognition
M6	Normalization	Component-level normalization
M7	Graph Construction	NetworkX DiGraph with 15+ edge types
M8	Confidence Annotation	Per-node OCR confidence scoring
M9	Embedding Generation	1024-dimensional embeddings with FAISS
M9b	Semantic Enrichment	Information type classification (14 types)
M10	Claim Extraction	Rule-based extraction of key claims
M11	Evidence Graph	Claim integration and relationship mapping
LLM-K	Knowledge Store	Direct LLM document comprehension
M12	Query Interpretation	Intent detection and decomposition
M13-15	Retrieval Pipeline	Bi-encoder, graph traversal, cross-encoder
M16-19	Answer Generation	Evidence-bound LLM answer generation
ASL	Answer Selection	Dual-path fact extraction and scoring
M20	Formatter	Citation linking and confidence scoring
M21	Suggestions	Context-aware follow-up generation

4. Technology Stack

LMDIS leverages state-of-the-art technologies across the entire processing pipeline:

Layer	Technology	Purpose
Backend Framework	Python 3.13, FastAPI, Uvicorn	High-performance async API server
Embedding Model	BAAI bge-large-en-v1.5	1024-dimensional semantic embeddings
Cross Encoder	ms-marco-MiniLM-L-12-v2	Evidence re-ranking for precision retrieval
Vector Store	FAISS IndexFlatIP	Efficient similarity search
Graph Engine	NetworkX DiGraph	Knowledge graph construction and traversal
LLM Integration	OpenRouter, Google Gemini, Anthropic Claude	Multi-model LLM orchestration
OCR Engine	PaddleOCR 3.4 with PyMuPDF	Scanned document text extraction
PDF Parsing	pdfplumber, PyMuPDF	Lossless content extraction
Frontend	HTML/CSS/JavaScript	Modern responsive interface
Storage	JSON and Pickle serialization	Isolated document storage

5. Key Capabilities

Lossless Structural Parsing

Preserves document layout, hierarchy, text formatting, table structures, and visual elements exactly as they appear in the source document. Adaptive extraction handles digital PDFs through PyMuPDF and pdfplumber, while scanned documents are processed through PaddleOCR with layout preservation.

Knowledge Graph Construction

Documents are transformed into directed graphs using NetworkX with 15+ edge types capturing semantic relationships, structural hierarchies, and cross-references. This graph representation enables sophisticated reasoning, traceability, and evidence-based query resolution.

Semantic Retrieval Engine

Replaces brittle keyword matching with embedding-based similarity search using BAAI bge-large-en-v1.5 (1024-dimensional vectors) indexed through FAISS. The retrieval pipeline combines bi-encoder search, graph traversal, and cross-encoder re-ranking for high-precision evidence extraction.

Evidence-Grounded Responses

Every answer is backed by verifiable source evidence with clear citations. The dual-path architecture generates answers from both the document knowledge graph and direct LLM reasoning, then arbitrates between them through a multi-signal Answer Selection Layer to ensure accuracy and prevent hallucinations.

Composite Confidence Scoring

Each citation carries a multi-factor confidence score blending semantic alignment (40%), OCR quality (30%), cross-encoder relevance (20%), and contribution verification (10%).

6. Citation Engine

The citation engine provides transparent and verifiable source attribution for every claim in the generated answer. Citations are constructed from the cross-encoder re-ranked evidence pipeline rather than simple keyword matching, ensuring high-precision source linking.

6.1 Multi-Signal Citation Attribution

Unigram Overlap (25%): Content word matching between generated answer and evidence text

Bigram Overlap (30%): Phrase-level attribution capturing multi-word concepts

High-Value Token Overlap (25%): Matching of numbers, proper nouns, and abbreviations

Retrieval Alignment Prior (20%): Score inherited from evidence retrieval pipeline

6.2 Citation Metadata

Every citation includes:

- Source text excerpt with page number reference
- Composite confidence score (0.0 to 1.0)
- Information type classification (fact, definition, statistic, methodology, result)
- Alignment score from the retrieval pipeline
- Contribution flag indicating independent attribution

7. API Reference

LMDIS exposes a RESTful API for programmatic access to all system capabilities:

Method	Endpoint	Description
GET	/	System health check and status
GET	/documents	List all uploaded documents
POST	/documents/upload	Upload and process PDF
GET	/documents/{id}/status	Check processing status
POST	/documents/{id}/query	Query with natural language
GET	/documents/{id}/suggestions	Get follow-up questions
GET	/documents/{id}/structure	Retrieve document structure

8. Performance Characteristics

8.1 Processing Time

Simple digital PDFs (10-50 pages): 30 to 90 seconds

Complex digital PDFs (50-200 pages): 2 to 5 minutes

Scanned documents (10-50 pages): 2 to 4 minutes

Large scanned documents (50-200 pages): 5 to 15 minutes

8.2 Query Response Time

Simple factual queries: 3 to 8 seconds

Complex analytical queries: 8 to 15 seconds

Multi-aspect queries with graph traversal: 10 to 20 seconds

8.3 Resource Utilization

Memory: 1.3 GB for embedding model + 50-200 MB per 100 pages

Storage: 10-50 MB per 100 pages depending on content

CPU: Multi-core systems show significant performance improvements

GPU: Optional CUDA acceleration for embeddings and re-ranking

9. Use Cases

Legal and Compliance Document Analysis

Analyze contracts, regulatory filings, and compliance documents with precise citation tracking. Extract obligations, deadlines, monetary terms, and compliance requirements with verifiable source attribution.

Financial Auditing and Reporting

Process financial statements, audit reports, and annual filings to extract financial metrics, performance indicators, and comparative analysis. Generate evidence-backed answers for audit queries with complete traceability.

Healthcare Record Processing

Analyze medical records, clinical trial reports, and research publications while maintaining data integrity. Extract patient information, treatment protocols, outcomes, and statistical findings with accurate citations.

Research and Academic Document Analysis

Process research papers, dissertations, and academic publications for literature review and knowledge synthesis. Extract methodologies, results, conclusions, and cross-references with academic rigor.

Enterprise Knowledge Management

Build searchable knowledge bases from corporate documentation, standard operating procedures, technical manuals, and policy documents. Enable employees to query organizational knowledge with source-verified answers.

10. Getting Started

10.1 Prerequisites

- Python 3.13 or higher
- Windows 10/11, Linux, or macOS
- Minimum 4 GB RAM
- 2 GB available disk space
- Internet connection for API calls and model downloads
- At least one LLM API key (OpenRouter, Google Gemini, or Anthropic Claude)

10.2 Installation

```
# Clone the repository
git clone https://github.com/nihcastics/LMDIS-with-Graph-Guided-Evidence-Aware-Reasoning.git
cd LMDIS-with-Graph-Guided-Evidence-Aware-Reasoning

# Create virtual environment
python -m venv .venv
.venv\Scripts\activate # Windows

# Install dependencies
pip install -r backend/requirements.txt
```

10.3 Configuration

Set environment variables for API keys:

```
# Windows (PowerShell)
$env:OPENROUTER_API_KEY="your_api_key_here"
$env:GOOGLE_API_KEY="your_api_key_here"
$env:ANTHROPIC_API_KEY="your_api_key_here"
```

10.4 Running the System

```
# Option 1: One-click launch (Windows) - Double-click start.bat
```

```
# Option 2: Manual launch
```

```
uvicorn backend.app.main:app --host 127.0.0.1 --port 8000
```

Access the frontend at: http://127.0.0.1:8000/frontend/index.html

11. Development Roadmap

Planned Features:

- **Multi-Document Reasoning:** Cross-document linking and comparative analysis
- **Performance Optimization:** Incremental processing and reduced memory footprint
- **Enhanced OCR:** Handwritten text recognition and complex layout detection
- **Real-Time Processing:** Streaming document processing and progressive query answering
- **Enterprise Features:** Role-based access control, audit logging, API rate limiting
- **Model Flexibility:** Pluggable embedding models and custom fine-tuning support

12. Limitations

The current implementation has the following known limitations:

- **Computational Requirements:** Large documents require significant processing time and memory
- **OCR Quality Dependency:** Accuracy depends on input document quality for scanned content
- **Single Document Scope:** Multi-document reasoning planned for future releases
- **Model Download Requirements:** 1.5 GB model weights require internet connectivity for first run

13. Contact and Repository Information

13.1 Primary Contacts

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13.2 GitHub Repository

Repository:
https://github.com/nihcastics/LMDIS-with-Graph-Guided-Evidence-Aware-Reasoning

Issues: https://github.com/nihcastics/LMDIS-with-Graph-Guided-Evidence-Aware-Reasoning/issues

Discussions: https://github.com/nihcastics/LMDIS-with-Graph-Guided-Evidence-Aware-Reasoning/discussions

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14.2 Copyright Notice

14.3 Academic Use

LMDIS was developed as a capstone project demonstrating advanced document intelligence capabilities. Academic use, research applications, and educational deployments are encouraged.

End of Document

LMDIS: Graph Guided Evidence Aware Reasoning for Enterprise Document Intelligence
Built with precision. Grounded in evidence. Designed for trust.